Dispersion-Aware Finite-Time Conditional Optimization

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Abstract

We replace the worst-case max-radius parameter by exact staple Fréchet dispersion and obtain a strictly feasible low-dispersion conditional channel.

Let C_* minimize $\sum_i d(C, S_i)^2 = \mathcal{D}$. If $d(C, C_*) \le \delta$, Minkowski's inequality in ℓ^2 gives

$$\left(\sum_i d(C, S_i)^2 \right)^{1/2} \le \sqrt{\mathcal{D}} + \sqrt{m}\, \delta.$$

Theorem 1. *The finite-time coherent bound remains valid with effective radius* $\sigma = \sqrt{\mathcal{D}/m}$ *and denominator energy*

$$(\sqrt{\mathcal{D}} + \sqrt{m}\delta)^2 = m(\sigma + \delta)^2.$$

Its unique Haar optimizer therefore satisfies the Stage 28 stationarity equation. At the maximum certified nonlinear curvature radius, the executable calibration gives a nonempty strict low-dispersion budget.

Proof. The displayed inequality supplies the denominator restriction, while a target plaquette outside radius ρ supplies numerator energy at least ρ^2. Substitution gives exactly the previously certified one-variable objective with $\eta = \sigma$. □

The complementary event $\mathcal{D} > D_*$ has deterministic incident energy at least D_* for every link value and hence bare exponent $D_*/(2t)$. Converting this action charge to a normalized retained-state partition numerator is not part of the theorem.

Remark 1. *The low-dispersion result is conditional and finite. The high-dispersion measure, adaptive histories, topology, RG iteration, and mass gap remain open.*